Exercise 1. Let I be an interval. A function $f : I \to \mathbb{R}$ is said to satisfy a Hölder condition with exponent $\alpha > 0$ if there exists a constant $C > 0$ such that if $x, y \in I$ then

$$|f(x) - f(y)| \leq C|x - y|^{\alpha}.$$

(a) Prove that if $f : I \to \mathbb{R}$ satisfies a Hölder condition with exponent $\alpha > 0$, then f is uniformly continuous on I.

(b) Prove that if $f : I \to \mathbb{R}$ satisfies a Hölder condition with exponent $\alpha > 1$, then f is constant.

Proof. Let I be any segment.

(a) Assume, for the purpose of proof, $f : I \to \mathbb{R}$ satisfies a Hölder condition with $\alpha > 0$ exponent. By f's Hölder condition, $\exists C > 0\ \forall x, y \in I$ such that

$$|f(x) - f(y)| \leq C|x - y|^{\alpha}.$$

Fix $\epsilon > 0$ and $\delta = C^{-1/\alpha}\epsilon^{1/\alpha}$. If $x, y \in I$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| \leq C|x - y|^{\alpha} < C\delta^{\alpha} = \epsilon.$$

Thus, f is uniformly continuous on I, as required of proof.

(b) Assume, for the purpose of proof, $f : I \to \mathbb{R}$ satisfies a Hölder condition with $\alpha > 1$ exponent. By f's Hölder condition, $\exists C > 0\ \forall x, y \in I$ such that

$$0 \leq |f(x) - f(y)| \leq C|x - y|^{\alpha} \implies 0 \leq \frac{|f(x) - f(y)|}{|x - y|} \leq C|x - y|^{\alpha - 1}.$$

Fix, without loss of generality, $x_0 \in I$. By the Squeeze Theorem of Real Functions,

$$0 \leq \left|\frac{f(x) - f(x_0)}{x - x_0}\right| \leq C|x - x_0|^{\alpha - 1} \implies |f('(x_0)| = 0 \implies f'(x_0) = 0.$$

Then f is differentiable on I. By the Derivative Characterization Theorem of Monotone Functions,

$$\forall x \in I : f(x) = 0 \implies \exists K \in \mathbb{R} : f(x) = K,$$

as needed of proof.

□

Exercise 2. Suppose $f : I \to \mathbb{R}$ is bounded, $g : I \to \mathbb{R}$ is differentiable at $c \in I$, and $g(c) = g'(c) = 0$. Show that $h(x) := f(x)g(x)$ is differentiable at c. *Hint*: You cannot apply the Product Rule.

Proof. Let $f : I \to \mathbb{R}$ be any bounded function. Suppose $g : I \to \mathbb{R}$ is any differentiable function at $c \in I$. Assume $g(c) = g'(c) = 0$. Define $h : I \to \mathbb{R}$ by $h(x) = f(x)g(x)$. We need to show that

$$h'(c) := \lim_{x \to c} \frac{h(x) - h(c)}{x - c} \in \mathbb{R}.$$

Given f is bounded on I, $\exists R \geq 0 \ \forall x \in I$ such that $|f(x)| \leq R$. By the Squeeze Theorem,

$$\begin{aligned} |h'(c)| := \lim_{x \to c} \left| \frac{h(x) - h(c)}{x - c} \right| &= \lim_{x \to c} \left| \frac{f(x)g(x) - f(c)g(c)}{x - c} \right| \\ &= \lim_{x \to c} \left| \frac{f(x)g(x)}{x - c} \right| = \lim_{x \to c} |f(x)| \left| \frac{g(x) - g(c)}{x - c} \right| \leq Rg'(c) = 0. \end{aligned}$$

Thus, $h'(c) = 0$, as needed. □

Exercise 3. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function. Prove that f is Lipschitz continuous if and only if f' is a bounded function.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ have a derivative everywhere. Assume, for the purpose of direct proof, $\exists K \geq 0 \ \forall x, y \in \mathbb{R}$ such that

$$|f(x) - f(y)| \leq K|x - y|.$$

Suppose, without any loss of generality, $x_0 \in \mathbb{R}$. If $x \in \mathbb{R}$, then

$$|f(x) - f(x_0)| \leq K|x - x_0| \implies \left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq K.$$

By the Order Relation of Convergent Functions,

$$|f'(x_0)| := \lim_{x \to x_0} \left| \frac{f(x) - f(x_0)}{x - x_0} \right| \leq K.$$

Thus, $\exists C \geq 0 \ \forall x \in \mathbb{R}$ for which

$$|f'(x)| \leq C,$$

as needed of direct proof. Assume, for the purpose of converse proof, $\exists K \geq 0 \ \forall x \in \mathbb{R}$ such that

$$|f'(x)| \leq K.$$

Suppose, without any loss of generality, $x < y$. Given f is differentiable on $\mathbb{R}$, f is continuous on $[x, y]$. By the Mean Value Theorem, $\exists c \in (x, y)$ such that

$$|f(x) - f(y)| = |f'(c)(x - y)| = |f'(c)||x - y| \leq K|x - y|.$$

Thus, $\exists C \geq 0 \ \forall x, y \in \mathbb{R}$ for which

$$|f(x) - f(y)| \leq C|x - y|,$$

as needed of converse proof. □

Exercise 4. (L'Hôpitial's Rule) Suppose $f : (a, b) \to \mathbb{R}$ and $g : (a, b) \to \mathbb{R}$ are differentiable functions and $c \in (a, b)$. Suppose that $f(c) = 0$, $g(c) = 0$, $g'(x) \neq 0$ when $x \neq c$, and that the limit of $f'(x)/g'(x)$ as x goes to c exists. Show that

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}.$$

Proof. Let $f : (a, b) \to \mathbb{R}$ and $g : (a, b) \to \mathbb{R}$ have derivatives everywhere. Suppose $c \in (a, b)$, $f(c) = g(c) = 0$ and $g'(x) \neq 0$ $\forall x \neq c$. Assume, for the purpose of proof,

$$\lim_{x\to c} \frac{f'(x)}{g'(x)} \in \mathbb{R}.$$

By the Mean Value Theorem, $\forall x_n \in (a, b) \setminus \{c\} \to c$ $\exists y_n, z_n \in (x_n, c) \cup (c, x_n) \to c$ such that

$$f(x_n) - f(c) = f'(y_n)(x_n - c) \ \wedge \ g(x_n) - g(c) = g'(z_n)(x_n - c)$$

and

$$\lim_{n\to\infty} \frac{f(x_n)}{g(x_n)} = \lim_{n\to\infty} \frac{f(x_n) - f(c)}{g(x_n) - g(c)} = \lim_{n\to\infty} \frac{f'(y_n)(x_n - c)}{g'(z_n)(x_n - c)} = \lim_{n\to\infty} \frac{f'(y_n)}{g'(z_n)} = \lim_{n\to\infty} \frac{f'(x_n)}{g'(x_n)}.$$

Thus,

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)} \in \mathbb{R},$$

as needed. □

Exercise 5.

(a) Suppose $f : [a, b) \to \mathbb{R}$ is continuous, differentiable in (a, b), and $f'(x) \to \lambda$ as $x \to a^+$. Prove that f is differentiable at a and $f'(a) = \lambda$.

(b) Suppose $f : (a, b] \to \mathbb{R}$ is continuous, differentiable in (a, b), and $f'(x) \to \lambda$ as $x \to b^-$. Prove that f is differentiable at b and $f'(b) = \lambda$.

(c) Suppose $f : (a, b) \to \mathbb{R}$ is continuous, and suppose f is differentiable everywhere except at $c \in (a, b)$ and $f'(x) \to \lambda$ as $x \to c$. Prove that f is differentiable at c and $f'(c) = \lambda$.

Proof. Start!

(a) Let $f : [a, b) \to \mathbb{R}$ be a differentiable function on (a, b). Suppose $f'(x) \to \lambda$ as $x \to a^+$. By the Mean Value Theorem, $\forall x_n \in (a, b) \to a^+$ $\exists y_n \in (a, x_n) \to a^+$ such that

$$f(x_n) - f(a) = f'(y_n)(x_n - a)$$

and

$$\lim_{n\to\infty} \frac{f(x_n) - f(a)}{x_n - a} = \lim_{n\to\infty} \frac{f'(y_n)(x_n - a)}{(x_n - a)} = \lim_{n\to\infty} f'(y_n) = \lim_{n\to\infty} f'(x_n).$$

Thus,

$$f'(a) := \lim_{x\to a^+} \frac{f(x) - f(a)}{x - a} = \lim_{x\to a^+} f'(x) = \lambda,$$

as needed.

(b) Let $f : (a, b] \to \mathbb{R}$ be a differentiable function on (a, b). Suppose $f'(x) \to \lambda$ as $x \to b^-$. By the Mean Value Theorem, $\forall x_n \in (a, b) \to b^- \; \exists z_n \in (x_n, b) \to b^-$ such that

$$f(x_n) - f(b) = f'(z_n)(x_n - b)$$

and

$$\lim_{n\to\infty} \frac{f(x_n) - f(b)}{x_n - b} = \lim_{n\to\infty} \frac{f'(z_n)(x_n - b)}{(x_n - b)} = \lim_{n\to\infty} f'(z_n) = \lim_{n\to\infty} f'(x_n).$$

Thus,

$$f'(b) := \lim_{x\to b^-} \frac{f(x) - f(b)}{x - b} = \lim_{x\to b^-} f'(x) = \lambda,$$

as needed.

(c) Let $f : (a, b) \to \mathbb{R}$ be a differentiable function on $(a, b) \setminus \{c\}$. Suppose $f'(x) \to \lambda$ as $x \to c$. By the Mean Value Theorem, $\forall x_n \in (a, b) \setminus \{c\} \to c \; \exists t_n \in (x_n, c) \cup (c, x_n) \to c$ such that

$$f(x) - f(c) = f'(t_n)(x_n - c)$$

and

$$\lim_{n\to\infty} \frac{f(x_n) - f(c)}{x_n - c} = \lim_{n\to\infty} \frac{f'(t_n)(x_n - c)}{(x_n - c)} = \lim_{n\to\infty} f'(t_n) = \lim_{n\to\infty} f'(x_n).$$

Thus,

$$f'(c) = \lim_{x\to c} \frac{f(x) - f(c)}{x - c} = \lim_{x\to c} f'(x) = \lambda,$$

as needed.

□